

Fig. 2: Combined actual-size PCB of the fan driver and the fan status monitor circuit

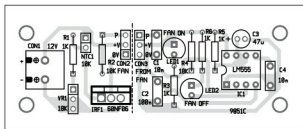


Fig. 3: Component layout of the PCB

PARTS LIST

Semiconductors:

- IC1 - LM555 timer
- IRF1 - 60NF06 MOSFET
- LED1, LED2 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3, R5, R6 - 1-k Ω -ohm
- R2, R4 - 10-k Ω -ohm
- VR1 - 10-k Ω -ohm potentiometer

Capacitors:

- C1, C4 - 10nF ceramic disk
- C2 - 100nF ceramic disk
- C3 - 47 μ F, 35V electrolytic

Miscellaneous:

- CON1 - 2-pin connector terminal
- CON2, CON3 - 2-pin connector
- NTC1 - 10-k Ω -ohm NTC thermistor
- VR1 - 12V regulated power supply
- 8-pin IC base

can be ignored.

The fan status monitor circuit is shown on the right side of the dotted line in Fig. 1. It is built around timer LM555 (IC1) and a few other components.

It is an add-on circuit for use with a 3-wire fan to indicate the running status of the cooling fan under control. The circuit is nothing but a minuscule visual indicator unit built around timer IC LM555 (IC1).

Green indicator LED1 in the circuit lights up when the fan driver is in active state, while red indicator LED2 lights up when it is in idle state. IC1 can determine whether the cooling fan is actually running or not. If both indicators (LED1 and LED2) are in on state, credibly it is a prime

indicator of cooling fan failure.

Here, IC1 is used as the fan pulse failure detector. IC1 is configured as a mono-stable multi-vibrator to monitor the rpm signal (tacho) output of the fan. If the fan stops in any case, red indicator LED2 will change from off to on. At the same time, output of IC1 changes from high (fan on) to low (fan off).

Since the circuit is one-shot, which is continuously retriggered from the input pulse, RC component values (R5-C3) can be changed to fit your needs. You should select RC components such that the delay is 30 per cent to 40 per cent higher than the period of input pulses.

In a 3-wire fan, first two wires (red and black) are for the power supply (as usual) and the third wire comes directly from the output of the built-in Hall sensor. This output generates tacho pulses in tune with the rotation of the fan. Often, a yellow-, green-, white- or blue-coloured wire is used as the tacho pulse wire in a 3-wire fan.

The combined actual-size, single-side PCB of the fan driver circuit and the fan status monitor circuit is shown in Fig. 2 and its component layout in Fig. 3. If needed, the PCB can be cut into two portions along the dotted line and mounted separately.

Construction and testing

Enclose the PCB in a separate suitable small box in such a way that the potentiometer can connect to the front panel easily. Ensure proper wiring to avoid any mistake. ●



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